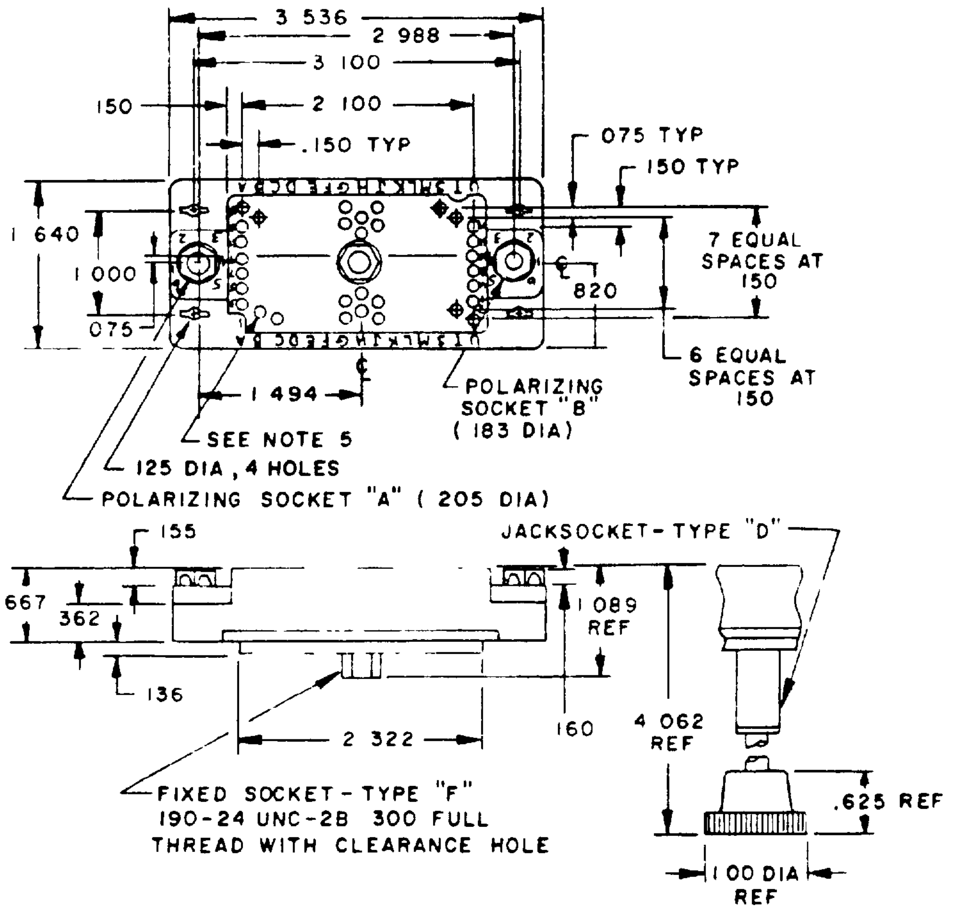
MIL-C-28731/4E
4 JUNE 1985
~~SUPERSEDING~~
MIL-C-28731/4D
25 May 1984

MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, INSERT (INSULATOR), FEMALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 100 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense

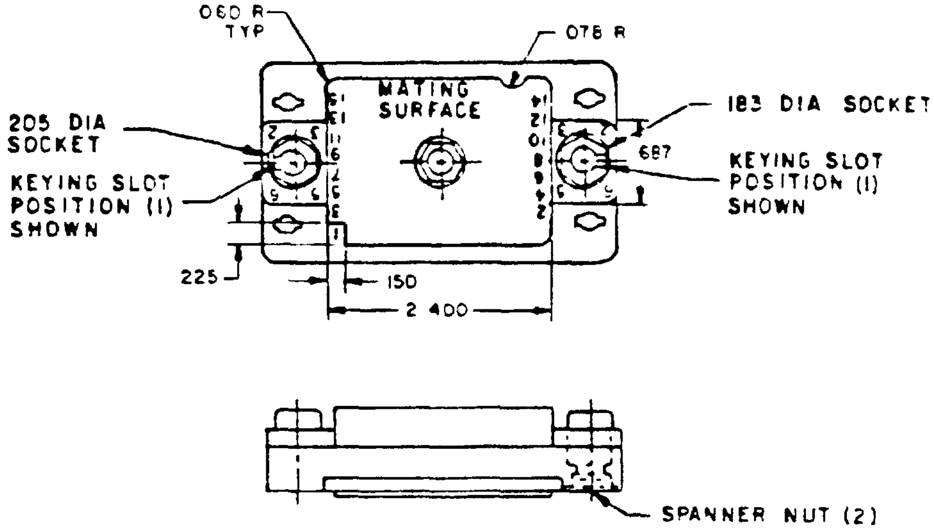
The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.


3 536
2 988
3 100
150
2 100
.150 TYP
075 TYP
150 TYP
1 640
1 000
075
820
7 EQUAL SPACES AT 150
6 EQUAL SPACES AT 150
1 494
SEE NOTE 5
125 DIA, 4 HOLES
POLARIZING SOCKET "A" (205 DIA)
POLARIZING SOCKET "B" (183 DIA)
155
JACKSOCKET-TYPE "D"
667
362
136
1 089 REF
160
4 062 REF
2 322
.625 REF
1 00 DIA REF
FIXED SOCKET-TYPE "F" 190-24 UNC-2B 300 FULL THREAD WITH CLEARANCE HOLE

INCHES	MM
.075	1.91
.125	3.18
.136	3.45
.150	3.81
.155	3.94
.160	4.06
.183	4.65
.190	4.83
.205	5.21
.300	7.62
.362	9.19
.625	15.88
.667	16.94
.820	20.83
1.000	25.40
1.089	27.66
1.434	37.95
1.640	41.66
2.100	53.34
2.322	58.98
2.9	75.90
3.100	78.74
3.536	89.81
4.062	103.17

FIGURE 1. Connector, female rectangular.

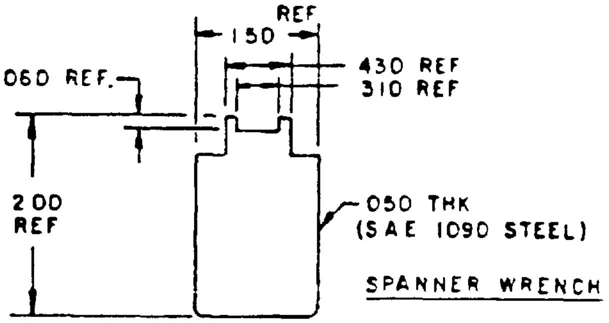
⊙ denotes changes


.060 R
TYP
.078 R
MATING
SURFACE
.205 DIA
SOCKET
KEYING SLOT
POSITION (1)
SHOWN
.183 DIA SOCKET
.687
KEYING SLOT
POSITION (1)
SHOWN
.225
.150
2.400
SPANNER NUT (2)

POLARIZING POSITION CHANGE

Keying position on both polarizing pins can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting pin in one of six positions then retightening spanner nut (clockwise).

NOTE Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below

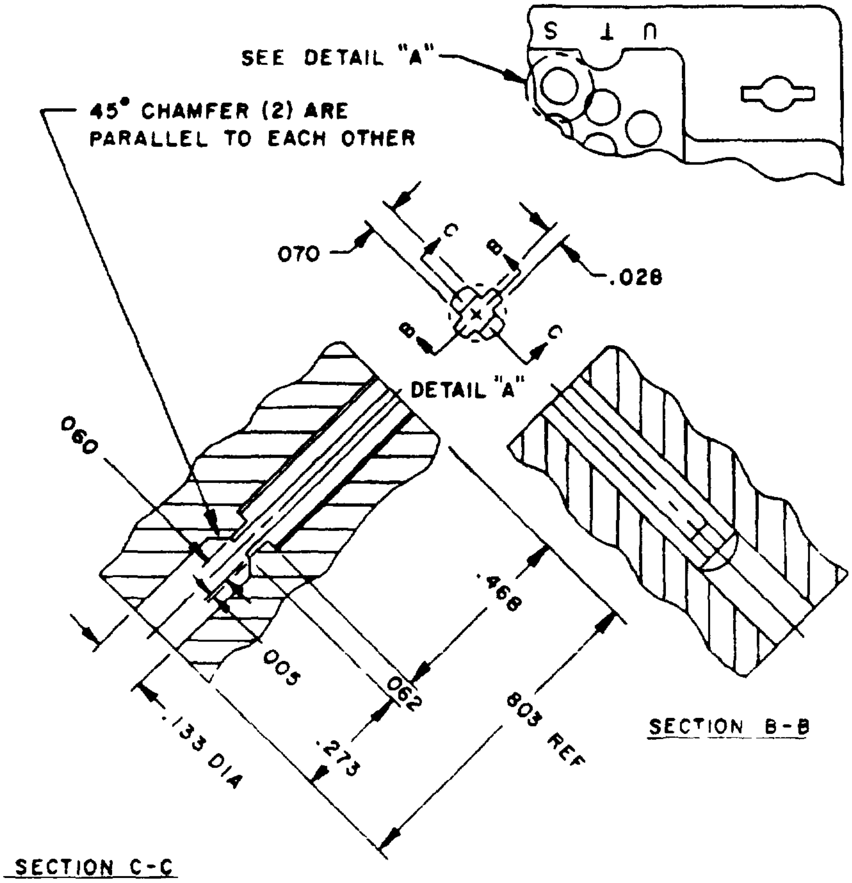

REF
.150
.060 REF.
.430 REF
.310 REF
2.00
REF
.050 THK
(SAE 1090 STEEL)
SPANNER WRENCH

Configuration optional

INCHES	MM
.050	1.27
.060	1.52
.078	1.98
.150	3.81
.165	4.19
.187	4.75
.225	5.71
.310	7.87
.430	10.92
.687	17.45
1.500	38.10
2.000	50.80
2.400	60.96

Configuration A - Connector polarization guide (layout)

FIGURE [illegible]. Connector, female rectangular - Continued



INCHES	MM
.005	0.13
.028	0.71
.060	1.52
.062	1.57
.070	1.78
.133	3.38
.273	6.93
.468	11.89
.803	20.40

Configuration B - Contact hole detail

NOTES
1. Dimensions are in inches
2 Metric equivalents are given for general information only
3. Tolerances between any two contact centers shall be ± .004 (0.10 mm)
4. Flatness, squareness, and parallelism shall be within dimensional tolerances
5 Contact hole details shall be in accordance with configuration B
6 Mating connector information
M28731/4-0073, -0075, and -0077 mate with M28731/3-0109, -0110, -0112 and -0113
M28731/4-0074 and -0076 mate with M28731/3-0111, -0114 and 0115
7 Polarization information
Polarization sockets are provided for indexing to preclude intermixing of adjacent connectors By using a standard type spanner wrench, polarization sockets can be rotated to desired position For parts supplied directly to the government polarization will be in the 1-1 position For other polarization positions see connector polarization guide configuration A (see marking, note 13d) For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment
8 Contact and terminal designations
Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1
9. Tolerance shall be ± .010 (0.25 mm) unless otherwise specified

FIGURE 1 Connector, female rectangular - Continued.

10 Patent notice
Patent No 3,248,686 owned by Elco Corporation The government has a royalty-free license under the above patent for the benefit of manufacturer of items called for in this specification, either for the government or for use in equipment to be delivered to the government

11. For direct government acquisition crimp contacts M28731/20-0001 are furnished in a plastic bag in each unit package (see marking, note 13a)

12 For direct government acquisition solder hole and wrappost contacts M28731/25-0001 and 26-0001 are factory installed in connector body (see marking, note 13a).

13 Marking
a Connectors supplied directly to the government shall be marked with the military part number (see table I). Example M28731/4-0074
b Connectors supplied with contacts to government end users shall be marked with the military part number (see table I) Example M28731/4-0074
c Connectors supplied without contacts to government end users shall be marked as follows Example M28731/4D (socket type from table I) The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles Example M28731/4D-0074
Ⓔ d Polarization position marking is not required unless otherwise specified When required polarization marking shall be in accordance with the connector polarization guide (see configuration A) The marking shall appear following the part number

14 Applicable contact insertion and removal tools

Insertion	MIL-I-81969/13-01
Removal	MIL-I-81969/11-01

15 When tooling with crimp contact MIL-C-28731 [illegible] is required for use with the connector, it shall be ordered separately

16 Rectangular shields shall conform to MIL-C-28731 [illegible] or [illegible]0004 and must be ordered separately

FIGURE 1. Connector, female, rectangular - Continued

REQUIREMENTS

Dimensions and configuration: See figure 1.

Mating and unmating forces (torque): 130 inch-ounces, maximum

Mechanical shock (specified pulse): Applicable

Part number: See table I.

TABLE I. Part number information.

Military part number	Actuating hardware	Polarizing socket position		Contact type
		A .205 Dia.	B .183 Dia.	
M28731/4-0073	Fixed socket Type F	1	1	Crimp MIL-C-28731/20 1/
M28731/4-0074	Jack socket Type J	1	1	Crimp MIL-C-28731/20 1/
M28731/4-0075	Fixed socket Type F	1	1	Solder hole MIL-C-28731/25 2/
M28731/4-0076	Jack socket Type J	1	1	Solder hole MIL-C-28731/25 2/
M28731/4-0077	Fixed socket Type F	1	1	Wrappost MIL-C-28731/26 2/

1/ See figure 1, note 11.
2/ See figure 1, note 12.

NOTE:
M28731/4-0073 supersedes M28731/4-0001 thru M28731/4-0036
M28731/4-0074 supersedes M28731/4-0037 thru M28731/4-0072

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, MI
Navy - AS
Air Force - 11, [illegible], 99
DLA - ES

User activities:
Army - AL, AV, [illegible]
Navy - MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-3488-3)